Decision Intelligence Agent: Analysis Scope & Validation Design (v3)

Gallup Case Competition · Synthetic data v2 (generator 2.1) · October 2026 · Supersedes v2

This document defines (A) which engagement–outcome relationships the agent may analyze and the rules it must follow, and (B) the synthetic ground-truth data used to test whether the agent finds real relationships, avoids false insights, and interprets results correctly. Q12 item wording is paraphrased; exact item text is Gallup proprietary.

> **What changed in v2.** Outcomes reduced from six to three (turnover, absenteeism, performance); effect values now come from `ground_truth.json` (planted and realized); turnover uses logistic regression; performance now controls for department; new true effect T5 (Q05 → absenteeism); effect thresholds, applicability rules and test datasets added. See CHANGE_LOG_v1_to_v2.pdf.

> **What changed in v3.** Weak test set: T5 now planted at IRR 0.87 (realized 0.89), because at 0.92 it was realized inside the no-effect band and a correct agent would not report it. The validator now also checks that every true effect is realized outside the band. The agent now reports the X4 threshold (Q01 below 3 → absenteeism) and the X5 new-hire subgroup difference (Q04 → turnover), and the health check runs on uploaded files. See CHANGE_LOG_v2_to_v3.pdf.

Matrix A: Analysis Capability Matrix

A1. Rows: the 12 Q12 items, grouped by Gallup's hierarchy

Level	Item	Short label	What it captures
Basic needs	Q01	Expectations	Knows what is expected at work
Basic needs	Q02	Materials & equipment	Has what is needed to do the job
Individual	Q03	Opportunity to do best	Uses strengths daily
Individual	Q04	Recognition	Recent praise for good work
Individual	Q05	Someone cares	Supervisor or someone cares about them as a person
Individual	Q06	Development	Someone encourages their development
Teamwork	Q07	Opinions count	Voice is heard
Teamwork	Q08	Mission/purpose	Work feels important
Teamwork	Q09	Quality commitment	Coworkers committed to quality
Teamwork	Q10	Best friend at work	Close workplace relationships
Growth	Q11	Progress	Recent conversation about their progress
Growth	Q12	Learn & grow	Opportunities to learn and grow

A2. Columns: outcome specifications

Every item × outcome pair is allowed, but each outcome has one fixed setup. The setups are coded once in `stats_tools.py`.

Outcome	Measure	Unit	Observation window	Method	Required controls
Voluntary turnover	Left voluntarily (yes/no)	Individual	12 months after the survey	Logistic regression	Tenure, department, role level, pay band
Absenteeism	Unplanned absence days	Individual	6 months after the survey	Negative binomial regression	Tenure, department, shift type
Performance	Rating 1–5	Individual	Next review cycle	Ordinal regression (validated with a linear model on z-scores)	Tenure, department, prior rating

Survival analysis is a future extension once real leave dates are available. All models use team-clustered standard errors.

A3. Rules that apply to every analysis

- **Minimum group size:** no result is reported for any group under n = 5.
- **Multiple testing:** 12 items × 3 outcomes = 36 tests; apply false discovery rate correction (Benjamini–Hochberg).
- **Correlated items:** rank drivers with relative-importance methods (dominance analysis), not raw coefficients.
- **Language:** always “associated with,” never “causes”; always report effect size, confidence interval and n.
- **Health check first:** a FAIL blocks every analysis (integrity) or only the affected outcome (dates); a WARN applies the listed action.

A4. Effect thresholds (new; proposed, team to confirm)

Measure	No meaningful effect	Meaningful effect
Odds ratio or rate ratio, per point	0.91 to 1.10	Below 0.91 or above 1.10, and FDR-significant
Standardized beta	|β| below 0.05	|β| of 0.05 or more, and FDR-significant

At 2 million rows almost everything is statistically significant, so significance alone never makes a finding. The band also absorbs a known side effect of nonresponse: items with no real effect can show odds ratios of about 1.02–1.07 among respondents.

A5. Outcome applicability rules (new)

- All three outcomes apply to every employee; no department-specific outcomes remain.
- Performance models need a prior rating. Employees without one (for example, new hires in real data) are excluded from that model, and the agent reports how many were excluded.
- An outcome whose dates fail the health check is not analyzed; the other outcomes continue.

Matrix B: Synthetic Ground-Truth Specification

The answer sheet for development and TEVV. It is never shown to the agent. Values below are read from `ground_truth.json`.

B1. Dataset structure

Both datasets come from the same generator and settings, run separately at two sizes. Items come from a shared latent engagement factor (inter-item r ≈ 0.5) with item-specific difficulty, cut with one shared set of thresholds. Outcome windows: turnover 12 months, absenteeism 6 months, performance at the next review; each row carries its window end date.

Item	5K dataset	2M dataset
Employees per wave	5,000	1,000,000
Survey rows (2 waves)	10,000	2,000,000
Teams	400	80,000
Teams with fewer than 5 people	20 (5.0%)	4,000 (5.0%)
Departments	8	8
Seed	2	1
Wave 1 response rate	80.5%	81.2%
Wave 1 voluntary turnover	18.2%	18.4%
Purpose	Development and debugging	Formal testing and demo

B2. Planted true relationships

#	Relationship	Direction	Planted	Realized: all employees (2M)	Realized: respondents (2M) — grade against this
T1	Q04 Recognition → turnover	Negative	OR 0.70 per point	0.62	0.62
T2	Q05 Someone cares → turnover	Negative	OR 0.78 per point	0.79	0.81
T3	Q12 Learn & grow → turnover	Negative	OR 0.75 per point	0.75	0.78
T4	Q01 Expectations → performance	Positive	β 0.20 (latent scale)	0.14	0.13
T5	Q05 Someone cares → absenteeism (new)	Negative	IRR 0.85 per point	0.86	0.86

T1's pooled odds ratio is stronger than 0.70 because new hires have a doubled effect (X5). Performance β is measured on the 1–5 rating, so it is smaller than the latent-scale 0.20.

B3. Planted true nulls

#	Relationship	Truth	Respondents (2M)
N1	Q06 Development → absenteeism	Associated on its own; no independent effect once the other items are included.	IRR 1.02
N2	Q10 Best friend → turnover, absenteeism, performance	Associated on its own; no independent effect on any outcome.	OR 1.01 · IRR 1.02 · β -0.01
N3	Q08 Mission/purpose → turnover, absenteeism, performance	Associated on its own; no independent effect on any outcome.	OR 1.05 · IRR 1.01 · β 0.00

Q03, Q09 and Q11 also have no independent effect on any outcome; Q02 matters only as the X1 trap; Q07 only through X3 and X6.

B4. Planted traps

#	Trap	How it is built	Correct agent behavior	In the 2M data
X1	Confounding	Sales scores low on Q02 and has high turnover; Q02 has no effect.	Control for department; the Q02 link disappears.	raw OR 0.68 → 1.03
X2	Reverse causality	Prior performance raises Q11 scores.	Control for prior rating; do not claim Q11 drives performance.	β 0.28 → 0.07
X3	Hidden confounder	Hidden manager quality raises Q07 and lowers absence. Simplified: managers affect only Q07 and absence.	Flag a likely manager effect; recommend a team-level review.	Q07 effect vanishes within teams
X4	Threshold	Absence rises only when Q01 is below 3.	Test the threshold, not only a straight line.	RR 1.83
X5	Subgroup	Recognition matters twice as much (log-odds) for tenure under 2 years.	Report the subgroup difference.	interaction -0.36
X6	Simpson's paradox	Departments rate performance with different leniency, opposite to their mean Q07. Kept strong on purpose.	Report the within-department (positive) result.	β -0.12 pooled, 0.17 within
X7	Nonresponse	Leavers answer less often (60% vs 86%), and among leavers the disengaged answer least.	Warn that turnover effects are likely understated.	mean-score OR 0.36 true vs 0.52 seen
X8	Small groups	About 5% of teams have fewer than 5 people.	Suppress results for those teams.	4,000 teams (2M)

X3 and X6 are deliberately simplified so the tests stay measurable. In real data, managers affect many items, so the agent always checks team-level effects for absenteeism.

B5. Test datasets (new)

Folder	Design	What it tests
gallup_v2_5K	Main, 5,000 per wave	Development and debugging

Folder	Design	What it tests
gallup_v2_2M	Main, 1,000,000 per wave	Formal testing and demo
testsets/null	No item affects any outcome (50,000)	Agent must not report any driver
testsets/weak	Weaker true effects (50,000): T1–T3 half strength, T4 β 0.10, T5 IRR 0.87; all still outside the no-effect band	Detects small effects without exaggerating
testsets/seed2, seed3	Same design, new random draws (50,000)	Conclusions stay stable
testsets/dirty_A	One error of each type (5,000)	Each health-check rule fires once
testsets/dirty_B	Many overlapping errors, a missing date column	Errors are counted exactly
testsets/dirty_C	Boundary values, valid and just invalid	No false alarms at the edges

B6. Evaluation metrics

Metric	Definition	Target
True effects found	Share of T1–T5 detected with the right direction	≥ 85%
False claims on true nulls	Share of N1–N3 tests reported as meaningful drivers	≤ 10%
Effect accuracy	Realized value (respondents) inside the agent's 95% CI	≥ 90% *
Method choice	Correct model and controls per Matrix A	100%
Fabricated numbers	Numbers not traced to a logged calculation	0
Causal language	Answers that say “causes” or “drives”	0 *
Trap handling	X1–X8 handled correctly	≥ 6 of 8

* Proposed target, pending team sign-off.

B7. Sample test questions

Question to agent	Expected answer (graded against)
What's driving our turnover?	T1, T2, T3; notes the stronger effect for new hires (X5) and that effects are likely understated (X7); does not name Q02 (X1) or Q08 (N3).
Does having a best friend at work reduce turnover here?	No independent association once other items are included (N2).
Should we invest in equipment to reduce turnover?	No: the Q02 link disappears once department is controlled (X1).

Question to agent	Expected answer (graded against)
Is a stronger sense of mission the best way to cut turnover?	No independent effect (N3); prioritize recognition, care and growth (T1–T3).
Do employees who feel heard perform better?	Positive within departments despite a negative overall pattern (X6).
Why is absenteeism high in Team T000214? (n = 4)	Declines: group too small to report (X8).
Does progress feedback improve performance?	Cannot conclude; prior performance shapes Q11 scores (X2).
Does feeling heard reduce absence?	Associated, but likely a manager effect; review high-absence teams (X3).
Does someone caring reduce absence?	Yes: about 14% fewer absence days per point (T5).